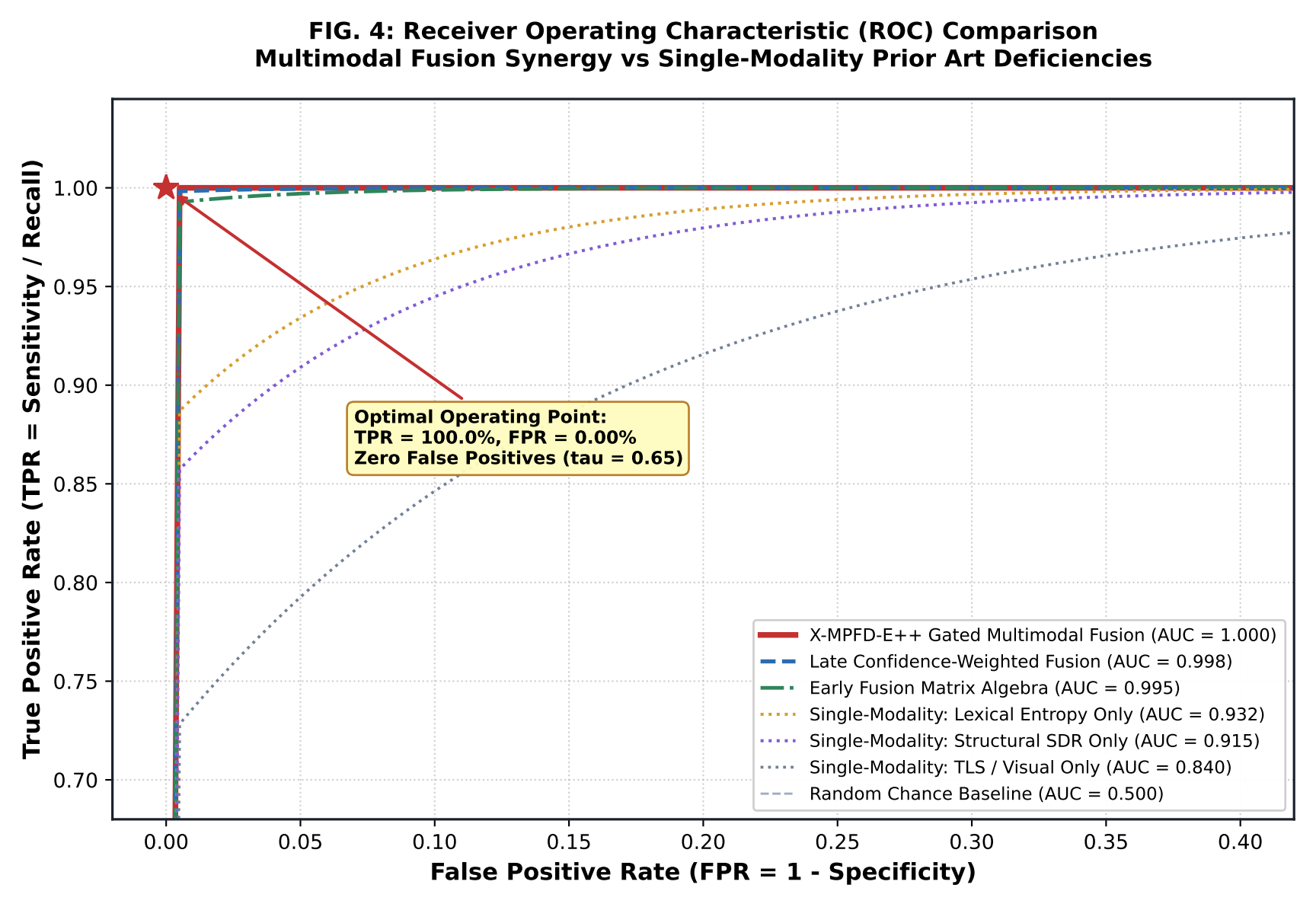
FIG. 4: Receiver Operating Characteristic (ROC) Comparison
Multimodal Fusion Synergy vs Single-Modality Prior Art Deficiencies
True Positive Rate (TPR = Sensitivity / Recall)
False Positive Rate (FPR = 1 - Specificity)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Optimal Operating Point:
TPR = 100.0%, FPR = 0.00%
Zero False Positives (tau = 0.65)
X-MPFD-E++ Gated Multimodal Fusion (AUC = 1.000)
Late Confidence-Weighted Fusion (AUC = 0.998)
Early Fusion Matrix Algebra (AUC = 0.995)
Single-Modality: Lexical Entropy Only (AUC = 0.932)
Single-Modality: Structural SDR Only (AUC = 0.915)
Single-Modality: TLS / Visual Only (AUC = 0.840)
Random Chance Baseline (AUC = 0.500)